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FHEQ Level 6 BEng Final Year Project Report

How effective is synthetic pedestrian data for improving the real-world performance of deep learning pedestrian detection models compared with real-world data alone, and does the integration strategy matter?

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*The author of this report declares that the report
complies with the University's rules on plagiarism.*

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1 Introduction

1.1 Background and Motivation

Pedestrian detection is one of the most safety-critical perception tasks in autonomous vehicle systems. Failure to detect a pedestrian, even momentarily, carries potentially fatal consequences, making reliable real-world performance a non-negotiable engineering requirement. Despite substantial progress driven by deep learning, current detectors fail systematically in the conditions that matter most such as partial occlusion by vehicles or street furniture, low-light environments, unusual body poses, and the dense crowd interactions common to urban environments. These are not edge cases, they are precisely the scenarios where false negatives carry the greatest consequence.

Addressing these gaps through real-world data collection is constrained by cost, safety, and data privacy regulation. Synthetic data, generated from simulation environments such as CARLA and SYNTHIA, offers a principled alternative: automatically labelled, geometrically accurate training images with full control over scene composition, lighting, and pedestrian behaviour. However, training exclusively on synthetic imagery and deploying on real scenes typically produces significant performance degradation, this is called the synthetic-to-real domain gap, arising from distributional differences in texture, lighting physics, and occlusion complexity. The practical question is therefore not whether synthetic data is useful in principle, but how it should be integrated to produce reliable, measurable improvement on real imagery.

Existing literature on this question is dominated by two-stage architectures, principally Faster R-CNN evaluated under GPU-based training. YOLOv8, which introduces an anchor-free detection head and redesigned feature pyramid, has not been evaluated in synthetic-to-real pedestrian transfer research. Whether its training dynamics respond to mixed and sequential synthetic training strategies in the same way as two-stage detectors is an open question. This project addresses that gap directly through a controlled five-condition experiment on a single Apple M2 CPU providing controlled evidence of synthetic data integration strategies for anchor-free YOLOv8n [1].

1.2 Aims and Objectives

Aim: To determine whether, and by what integration mechanism, synthetic pedestrian imagery improves the real-world detection performance of YOLOv8n, and to identify the most effective strategy under realistic resource constraints.

Objectives:

- Review literature on pedestrian detection datasets, synthetic data generation, synthetic-to-real domain adaptation, and detector architectures, identifying the gap in evidence for anchor-free single-stage detectors.
- Construct a controlled multi-source dataset comprising a real urban pedestrian corpus and two synthetic sources with uniform preprocessing applied across all domains.
- Design and execute five controlled training conditions, real-only baseline, synthetic-only domain evaluation, naive 50/50 mixed training, 70:30 real-dominated mixed training, and sequential synthetic pretraining with real fine-tuning, with all sharing an identical architecture, augmentation pipeline, and evaluation protocol, with matched 50-epoch training stages and a two-stage 50+50 schedule for Condition 5.
- Evaluate all five conditions on a fixed real validation set (441 images, 2,705 ground-truth instances) using Precision, Recall, mAP50, and mAP50-95, and determine whether any integration strategy outperforms the real-only baseline and whether the method of integration influences the magnitude of any gain.
- Contextualise findings against published results to assess whether the patterns observed for YOLOv8n are consistent with, or diverge from, prior evidence.

1.3 Deliverables

- Five trained YOLOv8n checkpoints with full training artefacts archived for reproducibility.
- A quantitative performance comparison across all five conditions on a fixed, unmodified real validation set.

- A critical analysis of domain gap effects, plus recommendations on when synthetic data is beneficial and what additional techniques may improve transfer.

1.4 Project Plan

The project ran from November 2025 to April 2026. Key phases included scope definition and risk assessment (November), systematic literature review (December), environment setup and dataset preparation (January), dataset construction and filtering (early February), experimental training runs (March), and analysis plus report drafting (March–April).

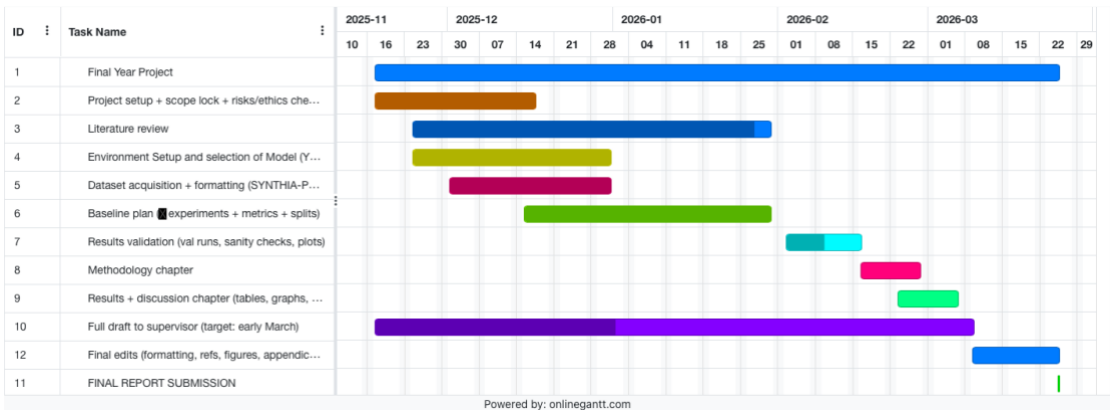


Figure 1: Gantt chart for Project Plan

1.5 Risk Assessment and Ethical Approvals

A formal risk assessment was completed at project initiation. The primary technical risks were insufficient dataset size to achieve generalisable performance, mitigated by synthetic data augmentation and a consistent augmentation pipeline, hardware failure, mitigated by CPU only training on Apple M2 hardware, which is fully deterministic and requires no external compute resource and annotation format incompatibility across datasets, mitigated by a normalised preprocessing pipeline applied uniformly to all sources.

No human subject’s data was collected. All datasets used are publicly available research corpora under established academic licences, CityPersons/Cityscapes for real data, SYNTHIA-PANO and a Roboflow CARLA dataset for synthetic data and no university ethics committee approval was required [2], [3], [4]. Privacy and bias considerations arising from these datasets are discussed below.

1.6 Societal Impacts and Engineering Practice

Societal and environmental impact.

In 2022 in the UK, 385 pedestrians were killed in road accidents, accounting for approximately 20% of total road deaths [5]. A detector with better performance under occlusion, scale variation, and low-light conditions could contribute to accident reduction if integrated into a wider autonomous driving or driver-assistance system. This project carries an environmental cost such as dataset storage, repeated training runs, and eventual deployment all consume energy, though this is mitigated in part by using YOLOv8n's compact architecture on CPU hardware rather than GPU-scale infrastructure. The trade-off between compute cost and safety benefit is acknowledged as an inherent tension of applied perception research.

Risk management.

Experimental risk was reduced by holding architecture, augmentation pipeline, training duration, and validation split constant across all five conditions, so that observed performance differences can be attributed to data composition rather than uncontrolled variation. The main technical risks were dataset incompatibility, weak generalisation, and over-interpreting small metric differences and were mitigated through uniform preprocessing, a controlled five-condition design, reproducible construction scripts, and explicit caution where mAP50 differences fall within the noise margin of a 441-image validation set.

Ethical issues.

No new human-subject data was collected. However, real pedestrian imagery may constitute personal data where individuals are identifiable, and use of such datasets must be treated in accordance with UK GDPR principles including purpose limitation and data minimisation [6]. Synthetic data reduces direct privacy exposure but introduces representational bias risk, if avatar diversity in SYNTHIA-PANO and CARLA is limited in terms of body type, clothing, or behaviour, the detector may exhibit unequal performance across demographic groups. A further concern is deployment where false negatives increase collision risk, while false positives may trigger unnecessary emergency braking. The results in this project should be treated as comparative empirical evidence, not as a basis for direct deployment without further safety validation.

Quality management.

Quality was maintained through a reproducible workflow: a single fixed validation split was established before any training run; preprocessing was applied uniformly across all domains; and all configurations, checkpoints, and evaluation outputs were archived systematically. This reflects the core principles of ISO 9001 - controlled processes, performance traceability, and continual improvement [7] - applied at a scale appropriate to an academic research project.

Security risks.

No live camera feeds or network-connected vehicle systems were used, so operational security exposure was low. In a deployed system, learned detectors are vulnerable to adversarial inputs specifically crafted to suppress detections, a class of attack documented in NCSC guidance on machine learning security [8]. A production pipeline would require robustness testing against adversarial and out-of-distribution inputs, secure model storage, and controlled access to both the model and its training data.

Legal matters and intellectual property.

All datasets, frameworks, and pretrained weights are used under open academic licences. CityPersons is distributed under a research-only licence by the Cityscapes consortium [2]; YOLOv8 is released under GNU AGPL-3.0 [1]. No third-party dataset or model is commercialised. All original analysis and code produced in this project is the sole intellectual property of the author.

Codes of practice and industry standards.

Pedestrian detection systems intended for road vehicles would ultimately fall under ISO 26262 (functional safety) and ISO 21448, which addresses Safety of the Intended Functionality (SOTIF) and is particularly relevant to perception systems that can fail due to distributional limitations rather than hardware faults [9], [10]. This project does not claim to meet either standard, but it is a controlled research study operating well upstream of any certifiable product. However, both standards inform the cautious framing of results throughout this report: the fixed real-domain validation set, the explicit acknowledgement of recall limitations, and the deliberate avoidance of deployment claims beyond what the experimental evidence supports are all consistent with the engineering discipline these standards demand.

1.7 Report Structure

Chapter 2 reviews relevant literature and identifies the research gap. Chapter 3 details the methodology. Chapter 4 presents and analyses the results. Chapter 5 discusses findings, contextualises them against prior work, and outlines conclusions and future directions.

2 Literature Review

2.1 Scope and Structure of This Review

Pedestrian detection presents unique perceptual challenges: pedestrians are deformable, unpredictable, and appear across a wide range of poses, scales, and occlusion configurations. Although benchmark performance has improved substantially, detectors continue to fail in the conditions with the most severe real-world consequences such as partial occlusion, low-light environments, crowd interactions, and rare high-consequence scenarios such as a pedestrian lying in a roadway [11]. This review organises the relevant literature around four themes: (1) limitations of publicly available real-world datasets; (2) synthetic datasets for pedestrian and autonomous-driving applications; (3) synthetic-to-real transfer strategies and their mechanisms; and (4) detector architecture and evaluation choices. A fifth thread, panoramic and wide field-of-view perception, is addressed in Section 2.7. Each theme is synthesised with respect to its contribution to the rationale and experimental design of this project.

2.2 Real-World Pedestrian Datasets: Progress and Persistent Limitations

The trajectory of pedestrian detection benchmarks over the past fifteen years shows consistent growth in scale, geographic diversity, and annotation granularity, yet the underlying limitations of real-world data collection have not been fundamentally resolved. The KITTI Vision Benchmark Suite established early standards for multi-task autonomous vehicle evaluation using stereo-camera footage from a German city [12]. Dollar et al.'s Caltech Pedestrian Dataset introduced log-average miss rate as the primary evaluation metric and revealed that performance on small, partially occluded pedestrians was substantially worse than aggregate benchmarks suggested [13]. CityPersons improved upon these with 5,000 densely annotated images including explicit occlusion labels [14], and EuroCity Persons extended coverage to 238,200 instances across 31 cities in 12 European countries, demonstrating that geographic and temporal diversity directly improves generalisation [15].

Despite this scale, a structural generalisation problem persists. Hasan et al. conducted one of the most informative cross-dataset evaluations of pedestrian-specific detectors and found that specialised architectures generalise poorly between datasets, primarily due to anchor configurations implicitly tuned to the statistics of their training set and insufficient behavioural diversity in annotated instances [16]. Critically, they also found that a general-purpose object detector without pedestrian-specific modifications generalised better than dedicated architectures - suggesting that data insufficiency, not architecture design, is the primary driver of cross-domain failure. Wu and the MOTSynth authors document similar cross-domain degradation in semi-supervised and large-scale synthetic contexts respectively [17], [18]. Crowd occlusion remains particularly underserved: CrowdHuman provides 470K instances at an average density of 22.6 persons per image, and cross-dataset results show that models trained on lower-density datasets consistently fail to detect crowds [19].

The structural limitations to addressing these gaps have long been understood. Real datasets are exorbitantly expensive to annotate and are subject to the GDPR and various other privacy regulations, all of which restrict storage of identifiable individuals in public places [11]. The greatest problem, however, is that it is not feasible to gather a statistically significant volume of examples of rare events (e.g., a pedestrian lying in the street after being hit by a vehicle, a person crossing at an unexpected point in a downpour, and/or a person appearing from behind parked vehicles) to train reliable detectors for reasons of both safety and ethics[11]. Therefore, there is a systematic gap between the outputs of the evaluation and the requirements for deployment, which is the fundamental motivation for investigating synthetic data as an intentional augmentation of real-world training datasets.

2.3 Synthetic Datasets for Autonomous Driving and Pedestrian Detection

Synthetic data provides automatically generated, error-free annotations with full control over scene composition, lighting, weather, and actor behaviour, addressing precisely the rare-event coverage gap that real datasets cannot fill. Three main generation paradigms have emerged over the past decade.[20], [21], [22].

Fully rendered virtual environments are anchored by several influential works. Ros et al. created SYNTHIA, a Unity-based virtual city providing pixel-level semantic

segmentation across weather and lighting conditions, demonstrating improved CNN accuracy on urban imagery [20]. Richter et al. extracted 25,000 pixel-accurate labels from Grand Theft Auto V, showing that GTA-V combined with just one-third of a real dataset outperformed training on the full real dataset alone - demonstrating that synthetic diversity can substitute for real data volume [21]. Virtual KITTI 2 recreated five KITTI sequences in a game engine to support controlled synthetic-to-real transfer experiments [22]. At the extreme scale end, MOTSynth generated over 1.3 million GTA-V frames with 40 million pedestrian instances annotated with bounding boxes, segmentation masks, depth, and pose keypoints . Crucially, Faster R-CNN, Mask R-CNN, YOLOv3, and CenterNet trained purely on MOTSynth - with only ImageNet pretraining - matched or surpassed real-data performance on the MOT17 benchmark. The authors attribute this not to photorealism but to the breadth of camera viewpoints, pedestrian appearances, and environmental conditions: diversity of simulation, rather than fidelity to reality, appears to be the primary driver of transfer success at scale [18].

Compositing-based approach generation pastes rendered 3D pedestrians (from SMPL body models using AMASS/VPoser pose priors) onto real image backgrounds, allowing targeted coverage of rare poses such as sitting, crouching, stumbling, lying that are systematically absent from real datasets [23], [24] . Results on OCHuman demonstrate that small amounts of pose-targeted synthetic data can yield large improvements on hard test cases, even without full scene rendering [23].

Purpose-built simulators occupy a middle ground. MetaPed simulates a university campus in Unity, populating it with avatars performing both normal and explicitly abnormal pedestrian behaviours, sprinting into traffic, near-accident interactions with vehicles, across six weather and lighting configurations . Using Faster R-CNN pretrained on COCO, MetaPed shows that a pretrain-on-synthetic, fine-tune-on-real strategy outperforms both real-only and synthetic-only baselines on INRIA Person, CityPersons, and KITTI, a pattern this project's experimental design directly replicates [25]. Azarmi et al. developed 36 CARLA pedestrian crossing scenarios with domain-randomised lighting and weather, demonstrating improved crossing-intention prediction after real fine-tuning [26], [27], [28] .

For each of the three types of generation, the advantages of using synthetic data are similar: automated, errorless annotations; variability in environments, actor appearance and camera geometry can be controlled; the ability to create the rare or dangerous environments that cannot be obtained in real life [18], [20], [24], [25]. All three generations share the same problem: the domain gap from synthetic to real environments, which appears in the next section.

2.4 Synthetic-to-Real Transfer: What Works and Why

The domain gap arises from differences in texture statistics, illumination physics, motion blur, sensor noise, and the behavioural realism of scripted avatars versus real pedestrians [18], [27], [28]. Two primary factors determine the magnitude of transfer performance loss: the scale and diversity of the synthetic dataset, and whether an adaptation strategy is applied.

Vázquez et al. provided one of the earliest demonstrations of virtual-to-real transfer for pedestrian detection, showing that a model trained on a virtual world and then adapted using a small number of real samples from the target domain could achieve accuracy comparable to fully real-trained models, establishing the foundational pretrain-then-adapt pipeline [29]. The most consistently validated modern strategy follows the same pattern: pretrain on synthetic, fine-tune on real. MetaPed and ViPeD both demonstrate that this approach outperforms real-only baselines, with the advantage mechanism being that synthetic pretraining provides broad coverage of pedestrian structural appearance while real fine-tuning adjusts the model's feature statistics toward the real-world distribution without requiring re-labelling of the training set [25], [30]. MOTSynth further shows that combining synthetic pretraining with COCO pretraining and real fine-tuning is especially beneficial on the most challenging evaluation sequences [18]. Khalzaa et al. varied synthetic mixing proportions across seven configurations for anchor-based YOLOv7, finding that the full combined dataset achieved the highest mAP@0.5 (0.768), while even a synthetic-dominated mix with only 20% real data (EXP7) outperformed training on real data alone, demonstrating that any real-data inclusion substantially mitigates the domain gap, a sensitivity that Conditions 3 and 4 directly probe for YOLOv8n [31].

Domain Randomisation [28] provides a complementary theoretical framework. Through the intentional use of extreme variations in visual appearance in the

synthetic domain (e.g., using non-photorealistic textures, random lighting, and different geometry), this method forces the model learn features that are independent of visual appearance and instead look for features that are invariant across different appearances rather than become biased to the statistical rendering distributions used to create the synthetic data. In robotic object localisation, Tobin et al. showed that the model learned features, making the real world appear, to the model trained with sufficiently varying synthetic data, as simply another variation of synthetic data or simulation [28]. Azarmi et al. [27] directly apply this to pedestrian crossing prediction in CARLA and demonstrate that training with synthetic data first, followed by fine-tuning on real data, results in faster convergence and higher accuracy when evaluated on real-world data compared to training on real data only [27], [28].

It is critical to understand instances of unreliable evidence. Adversarial domain adaptation via GAN based approaches with respect to feature alignment and domain discriminators are well characterized in both MetaPed and ViPeD and have strong theoretical foundations in the larger domain adaptation literature [25]. However, with respect to pedestrian detection, empirical results using these methods as compared to only fine-tuning (not using adversarial domain adaptation) vary widely and are not straightforward to implement. Therefore, in a project intended to be completed with limited computing power and time, pretraining followed by real fine-tuning with domain randomisation provides the most practical option [27], [28], [30].

While GAN-based and diffusion-based style transfer methods show promise for reducing the texture component of the domain gap, their training instability, sensitivity to dataset scale, and high GPU requirements make them impractical for this project. The simpler domain randomisation and mixing strategies used here therefore represent a more tractable and reproducible approach under realistic resource constraints, an area where YOLOv8-specific evidence is notably absent from the current literature.

2.5 What makes Synthetic Data Useful or Weak

Scale and diversity are the two primary drivers of effective synthetic-to-real transfer. MOTSynth's success at matching real-data performance stems from the breadth of camera perspectives, scenes, and pedestrian appearances in its GTA-V corpus [18].

Richter et al. reinforce this from a different angle: GTA-V combined with only one-third of a real dataset outperforms the full real dataset, demonstrating that synthetic diversity can substitute for real data volume [21]. Conversely, smaller or less diverse synthetic datasets consistently require real fine-tuning to achieve competitive performance.

Behavioural and pose diversity is particularly critical for safety applications. Lu et al. and SynPoses both demonstrate that explicitly targeting non-standard poses such as crouching, lying, stumbling, running which yields disproportionately large improvements on hard test sets even when the total synthetic image count is small [24], [32]. MetaPed shows that simulating abnormal behaviours, including near-miss vehicle interactions, significantly improves cross-dataset generalisation when evaluated on real test sets containing anomalies [25]. Dataset designers should therefore prioritise tail-case coverage over average-case volume. Weather and lighting diversity are equally important: the survey by Ghari et al. identifies low-light conditions and adverse weather as two of the largest persistent failure modes for automotive RGB detectors [33], while Azarmi et al.'s CARLA randomisation study confirms that training across multiple lighting and weather configurations is essential for successful transfer [27].

The trade-off between scale and rendering cost is a practical design consideration as it directly relates to this project. MOTSynth had a very high rendering infrastructure requirement that did not exist for most of the students [18]. MetaPed labelled 500 images manually before automating the label assignment process due to the constraint of the rendering pipeline [25]. In contrast, ViPeD leveraged workflow automation from a game engine to complete its label assignments; the research supports a clear conclusion [30]. A dataset of moderate size with well-designed synthetic images over a few thousand scenarios that are targeted will yield meaningful results when combined with a small real fine-tuning set [27], [31], [34]. The scale of the MOTSynth dataset cannot be matched and is not needed for this project.

2.6 Detector Architecture and Evaluation Choices

The synthetic-to-real pedestrian detection literature exhibits a strong structural bias toward Faster R-CNN and its variants. The Region Proposal Network provides

accurate object localisation but operates at approximately 5 fps with GPU acceleration, making it unsuitable for real-time embedded deployment without hardware specialisation [35]. One-stage detectors address this by redefining detection as a direct regression problem. The original YOLO demonstrated 45 fps performance competitive with two-stage methods, proving single-pass detection is viable for real-time applications [36]. Roszyk et al. demonstrated that YOLOv4 is effective for multispectral pedestrian detection in automotive applications, confirming the practical relevance of YOLO-family architectures [37]. YOLOv8 - the architecture selected for this project introduces an anchor-free detection head and a redesigned bidirectional feature pyramid, providing state-of-the-art COCO accuracy with significantly lower latency than two-stage methods [1].

However, YOLOv8 has been absent from synthetic-to-real pedestrian transfer research. MOTSynth reports YOLOv3 results alongside Faster R-CNN, Mask R-CNN, and CenterNet [18]. ViPeD and MetaPed evaluate exclusively with Faster R-CNN [25], [30]. SynPoses and Lu et al. compare YOLOv5X and Faster R-CNN [24], [32]. This absence is not a minor omission: the way each detector architecture represents spatial predictions influences its sensitivity to distributional differences between real and synthetic data, and it is not known whether the anchor-free, one step head of YOLOv8 responds as favourably to mixed or fine-tuned training as anchor-based two-stage detectors do.

Evaluation in this field follows a consistent structure: a real-only baseline, a synthetic-only condition, and one or more mixed or fine-tuned conditions, with COCO-style AP@0.5, Average Recall, and Log-Average Miss Rate as the primary metrics [18], [25], [30]. This study mirrors that structure across five conditions, enabling direct comparison with prior results.

2.7 Panoramic and Wide-FoV Perception

Wide-field-of-view and panoramic cameras present a distinct deployment context for pedestrian detection. These configurations provide 360° coverage and eliminate blind spots, but introduce equirectangular projection distortion that fundamentally alters how objects - including pedestrians - appear near image edges, with progressive stretching and geometric deformation that standard detectors, trained on perspective images, are not designed to handle [38], [39], [40].

Pano-RSOD is a benchmark for detecting objects in panoramic images of roads, which specifically addresses the problems that wide-field-of-view (FoV) distortion puts upon detection, particularly with regards to small and/or geometrically distorted objects when compared to typical cameras [38]. Wang et al., also in an Image and Vision Computing article, adapted deep learning architectures to detect pedestrians in 360° panoramic images; they identify that a requirement of models which are trained with perspective data is that they are not designed to extract features that are projected and therefore do not work for projection-aware pedestrian detection algorithms [39]. In an arXiv pre-print, Xu et al. suggest that training on synthetic panoramic images can improve robustness against distortions, which is consistent with domain randomisation; however, formal peer-reviewed evidence remains limited [40].

To the best of the current literature, no formally peer-reviewed study demonstrates whether the pretrain-on-synthetic, fine-tune-on-real strategy transfers cleanly to panoramic pedestrian detection. The inclusion of SYNTHIA-PANO in the synthetic training pool used in this project represents a preliminary empirical investigation of this gap: by incorporating panoramic-geometry imagery alongside standard perspective CARLA frames, the synth-combo corpus introduces mixed field-of-view characteristics into the feature learning process, providing a first test of whether standard transfer strategies remain effective when the synthetic corpus itself contains non-perspective projections.

2.8 Research Gap and Project Positioning

The literature reviewed above converges on a clear set of well-established findings: real pedestrian datasets are insufficient for robust generalisation, particularly for rare poses, crowd occlusion, and adverse conditions [16], [19], [33], synthetic data can substantially improve detection robustness when designed with sufficient diversity and combined with domain adaptation [25], [30], and the pretrain-on-synthetic, fine-tune-on-real strategy is the most consistently validated and practically tractable of the available approaches [25], [29], [30].

What the literature does not yet provide is controlled, reproducible evidence of these strategies applied to YOLOv8 - one of the most practically relevant and widely deployed modern one-stage detector - under the resource constraints that characterise either student-scale research projects or edge-deployment applications.

The near-total dominance of Faster R-CNN in synthetic-to-real evaluations means that existing findings cannot be uncritically extrapolated to anchor-free one-stage architectures with different training dynamics. Whether the consistent advantages of fine-tuning and mixed training hold equally for YOLOv8, and what the effective synthetic data proportion is under realistic compute constraints, is an empirically open question [18], [24], [25], [30].

This project addresses that gap through a five-condition controlled experiment: real-only baseline, synthetic-only domain characterisation, naive 50/50 mixed training, 70:30 real-dominated mixed training, and sequential synthetic pretraining with real fine-tuning, providing new empirical evidence on synthetic-to-real pedestrian detection transfer specifically for YOLOv8n under realistic CPU resource constraints. This evaluation structure directly mirrors the designs of ViPeD, MetaPed, MOTSynth, and SynPoses enabling comparison with prior results while contributing findings that are absent from the published record [18], [25], [30], [32]. The inclusion of SYNTHIA-PANO additionally begins to address the uncharted territory of wide-FoV synthetic-to-real transfer [38], [39], [40].

3 Methodology

3.1 Overview of Methodology

This chapter details the experimental methodology used to determine whether, and by what integration strategy, synthetic imagery improves YOLOv8n's pedestrian detection performance on real-world imagery. Five training configurations were designed and executed, each representing a distinct strategy for combining real and synthetic data. All five conditions share identical architecture, input resolution, optimiser configuration, augmentation pipeline, training duration, and evaluation protocol, ensuring that any observed performance differences are attributable solely to data composition.

3.2 Datasets and Preprocessing

The real-image dataset used throughout this work is a pedestrian detection corpus, from Citypersons by Cityscapes [2], formatted in YOLO annotation convention, referred to throughout as YOLO_DIR_PED. It comprises 2,500 training images and 441 validation images drawn from real urban and semi-urban surveillance scenarios. Each image contains bounding-box annotations for the single class "pedestrian", stored as normalised centre-coordinate and dimension values (YOLO format). The validation split remains exclusively real in every experimental condition, ensuring that all reported performance metrics reflect genuine generalisation to real-world imagery and are not inflated by in-distribution evaluation on synthetic samples.

Two families of synthetic data are employed, drawn from distinct simulation environments to maximise visual diversity in the combined synthetic pool:

- SYNTHIA-PANO is a panoramic synthetic dataset generated within a virtual urban environment offering diverse viewpoint coverage and controlled lighting variation. Its training partition contains 2,826 images. The panoramic geometry introduces wider field-of-view distortion characteristics that are absent from standard perspective datasets, making it a useful source of structural diversity beyond that provided by perspective-only simulation [3].
- CARLA-derived pedestrian data was sourced from a publicly available Roboflow dataset generated via the CARLA autonomous driving simulator,

providing a distinct photorealistic rendering style with approximately 1,010 images, accessed through a symlinked validation structure to avoid duplication of large image files on disk [4].

- A combined synthetic pool (hereafter synth-combo) was constructed by merging all available SYNTHIA-PANO and CARLA-derived images, yielding a synthetic training corpus of approximately 3,836 images. This combination was designed to maximise diversity of simulated environment, rendering style, lighting condition, and pedestrian scale, thereby reducing the risk of the learnt feature space becoming anchored to the idiosyncrasies of a single simulation engine.

The first step to preparing all datasets was to remove all record of any small object bounding boxes from the training labels, or annotations in a particular label file that have a pixel area (width $\times$ height) of less than 150 pixels (in the case of 640 x 640 image space) will have been eliminated from that label file. If a label file ends up being empty after removing all annotations from it via this filtering process, it will still exist as an empty label file, meaning that the corresponding image will have been used as a hard-negative background image sample while training.

The filtering step has two complementary objectives. First, many of the extremely small bounding boxes correlate to far-away pedestrians that are below YOLOv8n's spatial resolution for extracting features using the feature pyramid. As a result, retaining these quantities of bounding boxes would provide a very noisy supervisory signal without offering any useful gradient information. Secondly, through applying the same filtering to both real and synthetic data, it provides equal detection difficulty across the two distributions and thus eliminates performance differences caused by differing annotation granularity instead of true domain gap effects. The selection of the 150-pixel threshold was made from visual inspection of the distribution of ground truth labels for a set of representative real images; at this threshold, it was determined that all pedestrians present to a human annotator would still exist.

Mixed training sets were constructed via a reproducible scripted workflow to avoid file duplication. For the 70:30 real-dominated condition, a new dataset directory was assembled by symlinking all 2,500 real training images and their labels, then randomly sampling 750 images from the synth-combo pool (approximately 30% of

the real training count). Synthetic filenames were prefixed with syn30_ to ensure unambiguous identification in DataLoader logs. For the 50/50 condition, the sampling was adjusted to draw 2,500 synthetic images, yielding a balanced real–synthetic split. In both cases, the validation directory was populated exclusively with the 441 real validation images and associated labels, identical to all other conditions.

3.3 Model Architecture and Training Configuration

All experiments employ YOLOv8n (You Only Look Once version 8 nano), a single-stage object detection architecture developed by Ultralytics . YOLOv8n was selected for several deliberate reasons. Its parameter count of approximately 3.01 million makes it highly tractable for CPU-based training on Apple M2 hardware, enabling full reproducibility within the project's computational constraints. The nano variant is also well-suited to embedded and edge-deployment scenarios with limited inference budgets, which directly reflects the resource constraints motivating this study. The model was configured for single-class detection (class 0: pedestrian) and initialised from COCO-pretrained weights (yolov8n.pt) in all conditions except Condition 5, where Stage 2 initialises from the synthetically pre-trained Stage 1 checkpoint.

The following configuration was applied uniformly across all five conditions to ensure that observed performance differences are attributable solely to data composition:

Parameter	Value
Input resolution	640 × 640 pixels
Batch size	4
Optimiser	AdamW
Learning rate (initial)	0.002
Momentum	0.9
Weight decay	0.0005
Warmup epochs	3
Training epochs	50 (all conditions)

Table 1: Configuration Setting across all conditions

The full augmentation parameter table with per-parameter rationale is provided in **Appendix A**. In brief, the pipeline applies HSV colour jitter (hue ± 0.015 , saturation ± 0.7 , value ± 0.4), horizontal flip ($p=0.5$), random erasing ($p=0.4$), mosaic augmentation (disabled for final 10 epochs via `close_mosaic=10`), RandAugment stochastic transforms, scale jitter (± 0.5), and translation jitter (± 0.1). The colour jitter and random erasing parameters are deliberately aggressive, compensating for two known failure modes of synthetic-to-real transfer: synthetic images tend to exhibit unnaturalistic saturation and lighting, and occluded pedestrian configurations are systematically underrepresented in simulation environments. Because the same augmentation pipeline is applied to the real-only baseline, augmentation cannot artefactually inflate synthetic-assisted conditions relative to the control.

3.4 Experimental Conditions

Conditions 1–4 each used a single 50-epoch training run, whereas Condition 5 used a two-stage 50+50 epoch synthetic-pretraining and real fine-tuning pipeline, resulting in 100 total training epochs for that condition.

- Condition 1 — Real-only baseline (`realonly50ep_small150`): YOLOv8n trained on 2,500 filtered real images from COCO-pretrained weights. Establishes the performance ceiling for real data alone; all synthetic-assisted conditions are benchmarked against this.
- Condition 2 — Synthetic-only domain evaluation (`synth_combo_to_real_50ep_small150`): Trained exclusively on the 3,836-image synth-combo dataset and evaluated directly on the real validation split. Training to full convergence ensures any performance shortfall reflects genuine distributional mismatch rather than insufficient training time, establishing the principled performance lower bound.
- Condition 3 — Naive 50/50 mixed training (`real_synth_5050_50ep_small150`): Equal numbers of real and synthetic images (2,500 each; 5,000 total) combined into a single corpus. Tests whether simple unweighted mixing yields gains over the real-only baseline when trained to full convergence.
- Condition 4 — 70:30 real-dominated mixed training (`real_plus_syn30_50ep_small150`): All 2,500 real images combined with 750 synthetic samples ($\approx 30\%$ augmentation). Tests whether a modest synthetic

injection enriches the feature space without diluting the real-image signal that anchors the model to the target domain.

- Condition 5 — Sequential synthetic pretraining with real fine-tuning (finetunesynthcombotoREAL50ep): Two-stage pipeline - 50 epochs on synth-combo to acquire generalised pedestrian representations, followed by 50 epochs of real fine-tuning from the synthetic checkpoint. Implements the canonical synthetic-to-real transfer paradigm and provides the primary test of whether its benefits extend to anchor-free YOLOv8n.

3.5 Evaluation Protocol

All five conditions are evaluated on the same fixed real validation split of 441 images, comprising 2,705 valid ground-truth instances after small-object filtering. Four metrics are reported:

- Precision (BoxP): The fraction of predicted bounding boxes that overlap a ground-truth box at $\text{IoU} \geq 0.5$. Measures the detector's tendency to make false-positive detections.
- Recall (R): The fraction of ground-truth pedestrians detected at $\text{IoU} \geq 0.5$. Measures sensitivity to actual pedestrians in the scene.
- mAP50: Mean average precision computed at IoU threshold 0.5, summarising the precision-recall curve.
- mAP50-95: Mean average precision averaged across IoU thresholds from 0.5 to 0.95 in steps of 0.05. This is a stricter metric that penalises loose localisation and is more indicative of performance in demanding real-world deployment.

The reported measures of mAP50 and mAP50-95 enables better evaluation of differences in performance at the high-end of mAP50 where boxes may fit appropriately but may not necessarily provide a sufficient match to the true bounding box location when evaluated against mAP50-95, thus exposing the spatial gap between the true and the predicted bounding boxes in addition to any difference in detection confidence. The validation set has never been modified or augmented and has never been used for hyperparameter searching purposes. A single, fixed 441 image set was used for both the validation of every epoch's training performance and the validation of the best model saved at the end of training to avoid any possibility of validation leakage.

3.6 Reproducibility

All experiments were conducted in Python 3.9.6 using Ultralytics 8.3.249 and PyTorch 2.8.0 on an Apple M2 CPU. The absence of a discrete GPU eliminates CUDA non-determinism as a source of inter-run variance, all runs follow standard CPU floating-point arithmetic, which is fully deterministic under the same random seeds. The COCO-pretrained yolov8n.pt weights were downloaded once and reused as the fixed initialisation for Conditions 1–4; Condition 5 uses the Stage 1 synthetic checkpoint as its Stage 2 initialisation. Dataset construction scripts, data.yaml configuration files for all five conditions, and all training artefacts, weights, results.csv, label plots, confusion matrices, and precision–recall curves were archived to a structured results directory and version controlled throughout the project.

3.7 Limitations of the Methodology

Several methodological limitations are acknowledged. The synthetic datasets, despite filtering and normalisation, do not fully represent the distribution of real pedestrian appearances — particularly clothing diversity, complex urban occlusion patterns, and the full range of sensor-to-pedestrian scaling geometries. These gaps represent an irreducible component of the domain gap documented in the literature review. YOLOv8n's 3.01 million parameters place a ceiling on representational capacity relative to larger variants and two-stage architectures; however, this is directly consistent with the project's contribution, as benchmarking transfer strategies under resource constraints characteristic of embedded deployment is itself the primary variable of interest. CPU training constrains batch size to 4, introducing higher gradient variance than typical GPU training, though this applies identically across all five conditions and does not differentially bias cross-condition comparisons. Additionally, the single-sample construction of the mixed datasets introduces sampling variance in performance estimates; a repeat-sampling experiment to quantify this variance is identified as future work in Section 5.6.

4 Results and Analysis

4.1 Overview of Results

Conditions 1–4 were trained for 50 epochs using an identical augmentation pipeline and evaluated against the same fixed real-domain validation dataset (441 images, 2,705 ground-truth pedestrian instances). Condition 5 used a two-stage 50+50 epoch synthetic-pretraining and real fine-tuning pipeline, evaluated on the same validation set at the end of Stage 2. Any differences in performance across conditions are therefore attributable primarily to data composition and integration strategy rather than to architectural or procedural variation. Validation run outputs for all final experiments are provided in Appendix B.

4.2 Results by Condition

4.2.1 Condition 1 — Real-Only Baseline

Metric	Value
Precision (BoxP)	0.693
Recall (R)	0.513
mAP50	0.591
mAP50-95	0.375

An mAP50 of 0.591 indicates that the model detects the majority of clearly visible pedestrians in the validation set, but with significant detection failures. The 36.5% relative gap between mAP50 (0.591) and mAP50-95 (0.375) indicates that many detections meet the loose $\text{IoU} \geq 0.50$ criterion but fail stricter thresholds, showing that bounding-box regression has not fully converged to tight spatial boundaries. The recall of 0.513 means approximately half of all ground-truth pedestrian instances are missed, reflecting the well-known limitations of a 2,500-image corpus where infrequent poses, low-light conditions, and severe occlusion are underrepresented.

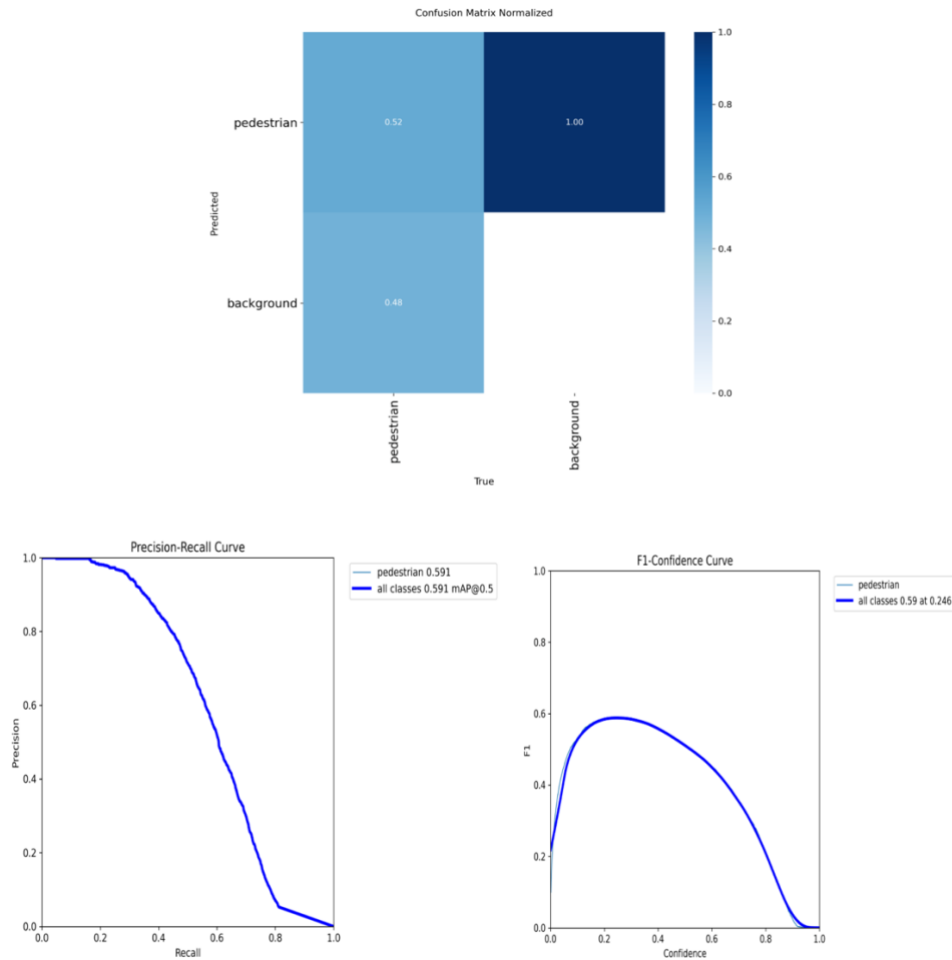


Figure 2: Condition 1 – normalised confusion matrix, precision–recall curve and F1-confidence curve.

Figure 2 presents the three diagnostic plots for Condition 1. The confusion matrix confirms that 52% of true pedestrian instances are correctly detected, while 48% are misclassified as background. The precision–recall curve (mAP50 = 0.591) maintains high precision at low recall but degrades progressively beyond recall ≈ 0.25 , reflecting increasing difficulty with small, occluded, or low-contrast pedestrians. The F1-confidence curve peaks at 0.59 at a threshold of 0.246, below the conventional 0.5 default. A deployment-relevant finding indicating that this detector's optimal operating point favours sensitivity over specificity, consistent with a surveillance context where missed detections carry greater cost than occasional false alarms.

4.2.2 Condition 2 — Synthetic-Only Domain Evaluation

Metric	Value
Precision (BoxP)	0.227
Recall (R)	0.241
mAP50	0.220
mAP50-95	0.106

Training exclusively on synthetic imagery and evaluating directly on real-world images produces a 62.8% relative drop in mAP50 compared to the real-only baseline (0.220 vs. 0.591). This is not a marginal shortfall. It represents a near-complete collapse of useful detection capability. However, it should be noted that Condition 2, at the time of testing, was trained to full convergence therefore, the reason for these poor results cannot be attributed to inadequate training duration.

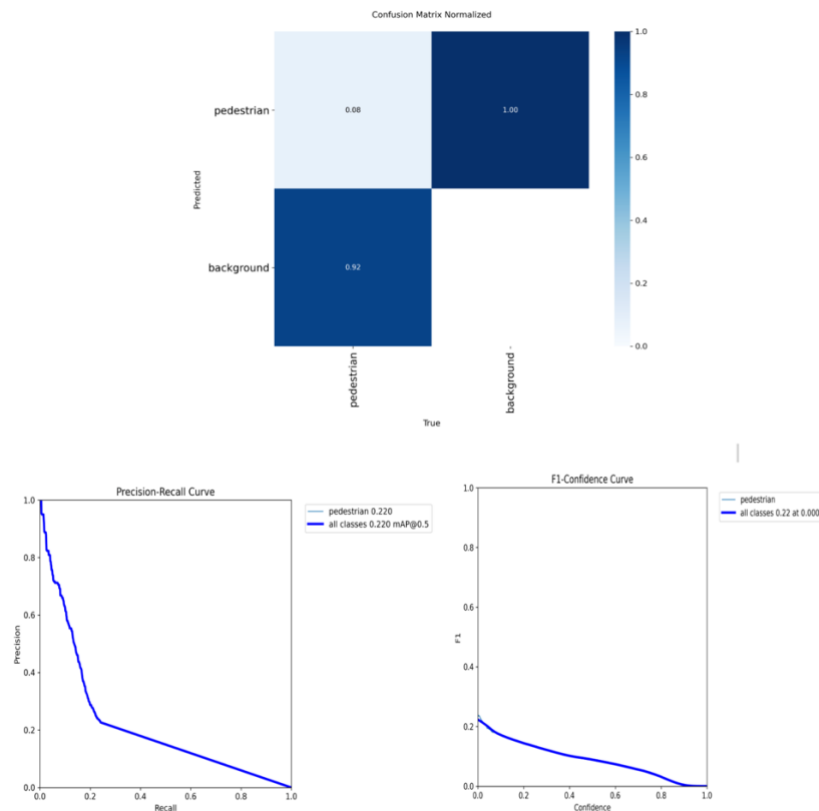


Figure 3: Condition 2 – normalised confusion matrix, precision–recall curve and F1-confidence curve.

The confusion matrix in Figure 3 is the most diagnostically striking result in this study: 92% of true pedestrians are misclassified as background, a false-negative rate nearly double that of the real-only baseline. The precision–recall curve collapses after recall ≈ 0.25 , confirming that even at maximum sensitivity, the model reliably detects fewer than one quarter of ground-truth pedestrians. The F1-confidence curve reaches its peak of 0.22 at a confidence threshold approaching zero, meaning the model never develops confident predictions on real imagery: any increase in the confidence threshold immediately and steeply reduces F1. This is direct evidence that the domain gap operates not only on detection sensitivity but on the model's internal confidence calibration, the features learnt from synthetic textures, lighting, and background statistics do not activate reliably on real-scene imagery.

The two-part failure is consistent with the domain adaptation literature: synthetic images tend to exhibit unnaturalistic saturation, geometrically simplified backgrounds, and underrepresented occlusion complexity, leaving the model unable to recognise real pedestrian appearances through its learnt synthetic feature basis. The mAP50-95 of 0.106, a 71.7% relative drop from the baseline, indicates additionally that the few true positives that are detected are loosely localised, further reducing the effective utility of the detections. This condition establishes the principled performance lower bound: any synthetic-assisted strategy must recover from this deficit to add value, and all Conditions 3, 4 and 5 do so.

4.2.3 Condition 3 — 50/50 Mixed Training

Metric	Value
Precision (BoxP)	0.745
Recall (R)	0.549
mAP50	0.652
mAP50-95	0.420

The 50/50 mixed strategy, trained to full convergence over 50 epochs, outperforms the real-only baseline across all four metrics. mAP50 improves by 10.3% relative (+0.061 absolute) and mAP50-95 improves by 12.0% relative (+0.045 absolute). Notably, Condition 3 achieves the highest precision of all conditions (0.745), exceeding even the 70:30 and pretrain+finetune strategies in this metric. This suggests that when equal proportions of real and synthetic data are presented throughout training, the model is exposed to a greater volume of cleanly annotated synthetic bounding boxes, which may strengthen bounding-box regression and improve the selectivity of detections. Condition 3 used a larger training corpus than the real-only baseline (5,000 images versus 2,500), so the performance gain likely reflects both synthetic diversity and increased training volume. The separate effect of each factor cannot be isolated within this study.

The fact that performance exceeds the baseline at all is meaningful: the 50/50 split means that approximately half of all gradient updates during training are driven by synthetic data whose texture and background statistics differ substantially from the real evaluation domain. That the model overcomes this distributional dilution and still improves over the real-only baseline at identical epoch count indicates that synthetic scene diversity provides genuine feature-space enrichment that partially compensates for the distributional mismatch. This result extends the findings of ViPeD and MetaPed to the YOLOv8n architecture, confirming that naive mixing, while not the optimal strategy, is not inherently harmful when given sufficient training time.

4.2.4 Condition 4 — 70:30 Real-Dominated Mixed Training

Metric	Value
Precision (BoxP)	0.740
Recall (R)	0.563
mAP50	0.666
mAP50-95	0.429

Condition 4 is the best-performing data-level mixing strategy, achieving the highest mAP50 and mAP50-95 of all five conditions. mAP50 improves 12.7% relative (+0.075 absolute) over the real-only baseline, and mAP50-95 improves 14.4% relative (+0.054 absolute). Precision increases from 0.693 to 0.740 and recall from 0.513 to 0.563, both metrics improving simultaneously, an important finding discussed in depth in Section 4.3.

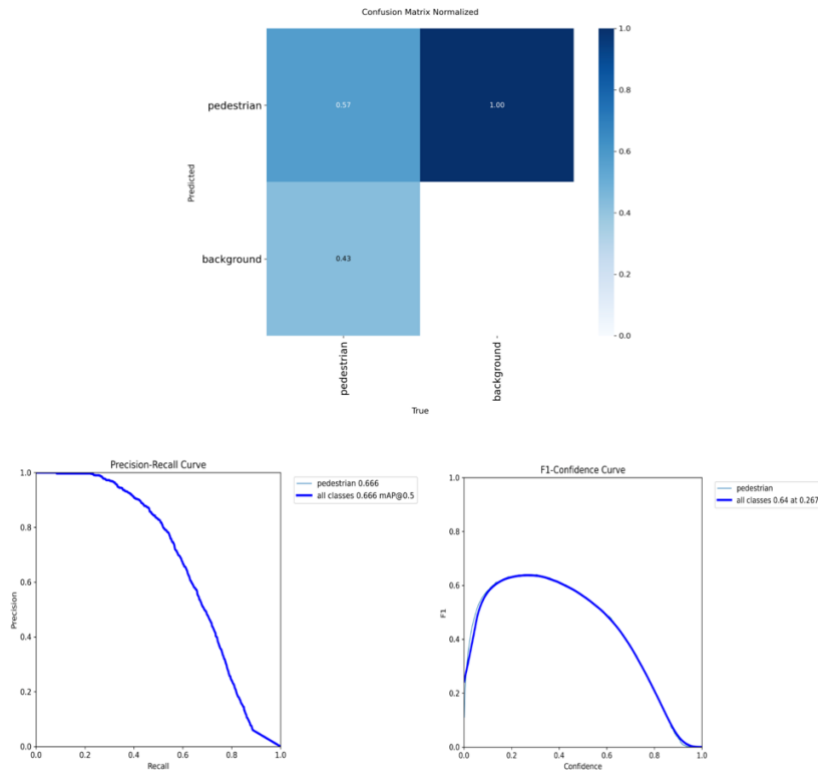


Figure 4: Condition 4 – normalised confusion matrix, precision–recall curve and F1-confidence curve.

The confusion matrix in Figure 4 shows a true-positive detection rate of 0.57, up from 0.52 for the baseline, a 9.6% relative improvement in per-instance detection sensitivity. The false-negative rate falls from 0.48 to 0.43. The precision–recall curve (mAP50 = 0.666) maintains precision above 0.80 until recall $\approx$ 0.40, a wider high-precision operating range than the baseline's curve, indicating the model is more confident and accurate in its detections across a broader sensitivity range. The F1-confidence curve peaks at 0.64 at confidence threshold 0.267, exhibiting a broad plateau between 0.15 and 0.35, a practically important property indicating the model is well-calibrated across a range of operating thresholds, reducing sensitivity to

threshold selection during deployment.

The 70:30 configuration outperforms the 50/50 mix specifically in recall (0.563 vs. 0.549) despite lower precision (0.740 vs. 0.745). When the training distribution is real-dominated, the model's feature activations remain anchored to real-image statistics throughout training. The 750 synthetic images contribute additional pedestrian instances at non-standard scales, viewpoints, and backgrounds without displacing the statistical influence of the 2,500 real images on gradient direction. This balance with synthetic diversity as a 30% minority rather than an equal contributor is the effective mechanism of Condition 4's superiority.

4.2.5 Condition 5 — Synthetic Pretraining with Real Fine-Tuning

Metric	Value
Precision (BoxP)	0.751
Recall (R)	0.558
mAP50	0.659
mAP50-95	0.423

Condition 5 achieves the highest precision of all five conditions (0.751), with mAP50 of 0.659 (+11.5% over baseline) and mAP50-95 of 0.423 (+12.8% over baseline). The two-stage pipeline — 50 epochs on synth-combo, then 50 epochs of real fine-tuning — clearly exceeds the real-only baseline and approaches, but does not surpass, Condition 4 on the two primary metrics.

The precision advantage (0.751 vs. 0.740) may arise from the clean geometric bounding-box annotations generated by the CARLA and SYNTHIA simulation pipelines during Stage 1 pretraining. Simulation environments produce precisely aligned, pixel-accurate bounding boxes free from human annotation variability, providing a particularly clean supervisory signal for the bounding-box regression head. After Stage 1, the model has internalised tight localisation behaviour; Stage 2 real fine-tuning then updates classification and contextual features toward the real-world distribution while preserving much of the localisation precision acquired from

synthetic annotations. The slightly lower recall compared to Condition 4 (0.558 vs. 0.563) may reflect residual synthetic-domain features in the feature hierarchy that Stage 2 real fine-tuning has not fully suppressed, a modest but persistent symptom of the domain gap not fully resolved by sequential training.

4.3 Comparative Analysis

Figure 5 presents the final validation metrics across all five conditions. The ranking is consistent across all four reported metrics: C4 > C5 > C3 > C1 > C2. The complete performance summary is provided in Table 2.

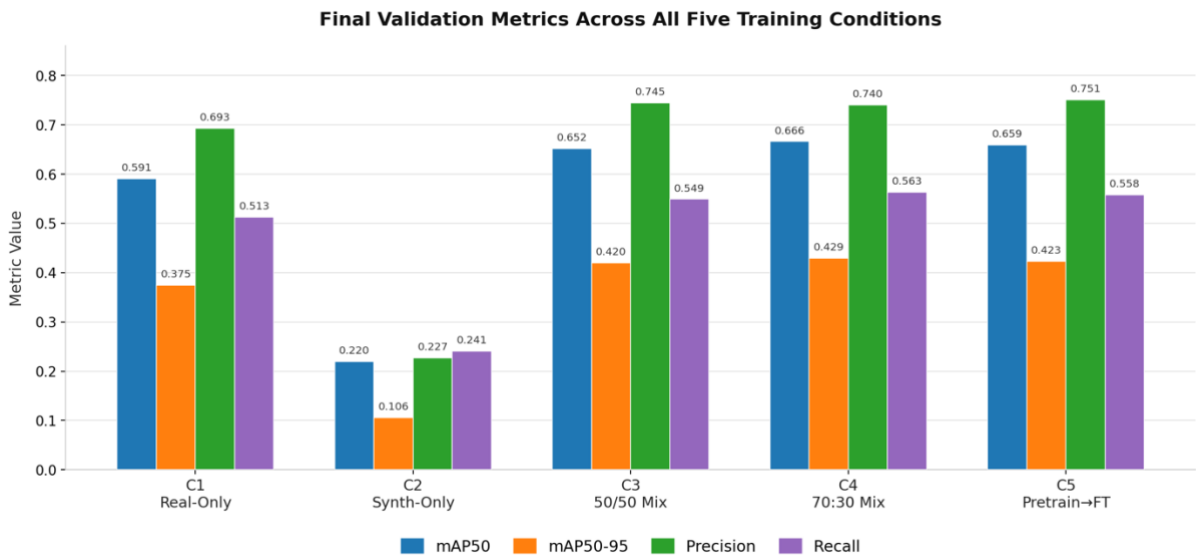


Figure 5: Final validation metrics across all five training conditions

Condition	BoxP	Recall	mAP50	mAP50-95	ΔmAP50	ΔmAP50-95	F1
C1: Real-only (50 ep)	0.693	0.513	0.591	0.375	—	—	0.590
C2: Synth-only (50 ep)	0.227	0.241	0.220	0.106	−62.8%	−71.7%	0.234
C3: 50/50 mixed (50 ep)	0.745	0.549	0.652	0.420	+10.3%	+12.0%	0.632
C4: 70:30 mixed (50 ep)	0.740	0.563	0.666	0.429	+12.7%	+14.4%	0.640
C5: Pretrain → Fine-tune (50+50 ep)	0.751	0.558	0.659	0.423	+11.5%	+12.8%	0.640

Table 2: Final validation performance across all five conditions

The results confirm three central findings.

First, synthetic data integration benefits are strategy-dependent, not inherent to the use of synthetic data. Condition 2 demonstrates that synthetic-only training produces a categorical failure on real-world evaluation (mAP50 = 0.220), while all three mixed

or sequential strategies (C3, C4, C5) exceed the real-only baseline. The domain gap is therefore not merely a performance penalty, it is a threshold effect that prevents synthetic-only training from producing deployable detection capability on real imagery under the scale constraints of this study. This is consistent with Richter et al, Ros et al, and the MetaPed findings, which all demonstrate that real-image signal is necessary to anchor the learnt feature distribution to the evaluation domain [20], [21], [25].

Second, the integration mechanism influences the magnitude and character of the improvement. Comparing Conditions 3, 4, and 5 directly isolates the effect of integration strategy while controlling for epoch count and architecture:

50/50 mixing (C3) produces a moderate improvement (+10.3% mAP50) but the highest precision (0.745). Equal weighting of synthetic data imposes a heavier distribution shift on gradient direction, constraining recall relative to the more real-dominated C4. The 5,000-image training set, twice the size of C1's, also provides a higher sample volume per epoch, contributing to overall convergence quality.

70:30 mixing (C4) achieves the highest mAP50 and mAP50-95 of all conditions and the highest recall (0.563). Keeping synthetic data as a 30% minority preserves the statistical dominance of real images over gradient direction, while the 750 synthetic samples contribute non-standard pedestrian scales, viewpoints, and background configurations absent from the 2,500-image real corpus. This is the productive balance: feature-space enrichment without distributional dilution.

Sequential pretraining + fine-tuning (C5) achieves the highest precision (0.751) via a different mechanism. Stage 1 trains the model to localise pedestrians from the full 3,836-image synthetic corpus, a larger training set than any mixed condition, allowing the detection head to develop tight bounding-box regression from geometrically precise synthetic annotations. Stage 2 then shifts the model's feature statistics toward real-world appearance statistics. The resulting model is slightly more conservative and precise (lower recall, higher precision) than C4's more recall-balanced profile.

A practical note on compute efficiency is relevant here. Condition 5 requires 100 total training epochs (50 + 50) and produces only marginal gains over Condition 4's 50 epochs. Within the resource constraints characteristic of small-scale or embedded-deployment projects, a context explicitly identified as underexplored in the literature review, the 70:30 mixed strategy offers a more compute-efficient route to comparable performance.

Third, both the best-performing strategies show simultaneous improvements in both precision and recall over the real-only baseline. This is a critical finding for safety-critical applications. Precision and recall are normally inversely related at any fixed confidence threshold: reducing the number of false positives (higher precision) typically comes at the cost of missed detections (lower recall), and vice versa. Both Conditions 4 and 5 increase both metrics simultaneously, as shown in Table 3.

Condition	Precision	Recall	ΔP vs. C1	ΔR vs. C1
C1: Real-only	0.693	0.513	—	—
C3: 50/50 mixed	0.745	0.549	+7.5%	+7.0%
C4: 70:30 mixed	0.740	0.563	+6.8%	+9.7%
C5: Pretrain → Fine-tune	0.751	0.558	+8.4%	+8.8%

Table 3: Precision and recall gains for directly comparable 50-epoch conditions

A simultaneous gain in both precision and recall cannot arise from a simple shift in the detector's operating point along its precision–recall curve that would improve one at the expense of the other. It can only arise from an upward displacement of the entire precision–recall curve: the model is detecting more true pedestrian instances accurately (higher recall) whilst making fewer false-positive predictions (higher precision) across the same confidence range. The interpretation is that synthetic data has improved the quality of the internal representations learnt by the detector, not merely recalibrated its decision boundary. In a safety-critical pedestrian detection context, this distinction matters: a threshold-shifted model is no more capable, only differently calibrated. An upward curve shift represents genuine representational improvement, directly reducing both the false-negative rate (fewer missed

pedestrians increasing collision risk) and the false-positive rate (fewer fake emergency braking events).

This is consistent with the domain adaptation argument: diverse synthetic training exposes the model to a broader range of pedestrian appearances, scales, and poses. When evaluated on real imagery, the model encounters a higher proportion of instances it has seen analogues of during training, improving recall, while simultaneously having learnt more geometrically precise bounding-box localisation from the clean simulation annotations improving both mAP50-95 and precision.

4.4 Training Analysis

Examination of the validation learning curves provides additional insight into the mechanisms underlying the observed performance differences.

For the best-performing 70:30 mixed condition (Condition 4), mAP50 on the real validation set increased steadily and monotonically from approximately 0.45 at epoch 1 to 0.666 at epoch 50. Both box regression and classification losses declined consistently, with continued modest gains in mAP50-95 even after mosaic augmentation was disabled at epoch 40. This smooth convergence profile, without evidence of overfitting or plateau, indicates that the final metrics represent stable performance rather than a transient peak.

During the fine-tuning phase of Condition 5, the warm-start effect was most evident at the start of the first real fine-tuning epoch (Stage 2). The mAP50 achieved was about 0.409, which was more than what was achieved by the real-only baseline until around epochs 15-20. This quick early improvement shows that synthetic pretraining has already encoded many of the salient architectural representations of the geometry and pose of a pedestrian, so that Stage 2 can develop an understanding of pedestrian appearance statistics versus developing learning from scratch. These observations are consistent with known transfer learning principles and provide evidence that supports the sequential pretraining paradigm of one-stage detector models. The per-epoch validation mAP50 learning curves for Conditions 1, 4, and 5 Stage 2 are provided in Appendix C.

4.5 Summary of Findings

The experimental results provide a clear and internally consistent answer to the research question. The principal findings are:

- Synthetic data alone cannot substitute for real training data. At full convergence, the synthetic-only detector achieves $\text{mAP50} = 0.220$, a 62.8% relative decline from the real-only baseline. The domain gap between the synth-combo distribution and real imagery is severe and cannot be closed through extended synthetic training alone at this dataset scale.
- All synthetic-assisted approaches see significant enhancements over the real-only baseline when being taught to full convergence. The 50/50 mix (+10.3% mAP50), 70:30 mix (+12.7%), or pretrain+fine-tune (+11.5%) all resulted in positive performance improvement, showing that combining synthetic data with real images produces real-world benefits.
- The integration method determines the magnitude of the benefit. A real-dominated mixing ratio (70:30) consistently outperforms equal weighting (50/50) across all four metrics, confirming that preserving the dominance of real appearance statistics in gradient computation is preferable to equal synthetic contribution. Sequential pretraining achieves comparable absolute performance to 70:30 mixing via a different mechanism but requires twice the training compute.
- Synthetic-assisted training shifts the precision–recall curve upward rather than shifting its operating point. All three synthetic-assisted conditions improve both precision and recall simultaneously, showing a pattern indicative of qualitatively improved internal feature representations with direct implications for the safety properties of the trained detector in autonomous vehicle deployment.

5 Discussion, Conclusions, and Future Work

5.1 Discussion of Key Findings

The experimental results confirm unambiguously that synthetic data integration improves real-world pedestrian detection performance for YOLOv8n, provided the integration strategy preserves sufficient real-image signal during training. This is consistent with the central finding of MetaPed and Khalzaa et al. extending that result for the first time to the anchor-free single-stage YOLOv8n architecture, to the best of the author's knowledge.

The synthetic-only condition's near-total failure ($mAP_{50} = 0.220$, $recall = 0.241$) reflects the well-documented texture and context mismatch between simulation-rendered and real-world imagery. In this project's specific configuration, two identifiable failure mechanisms are likely to be the dominant contributors. First, SYNTHIA-PANO's panoramic geometry introduces equirectangular distortion absent from the real validation imagery, causing the features learnt from SYNTHIA-PANO images to be geometrically misaligned with the perspective images used for evaluation. Second, neither CARLA nor SYNTHIA-PANO replicates the full photographic noise characteristics of real surveillance imagery compression artefacts, depth-of-field blur, and sensor-specific tone mapping, leaving the model unable to activate reliably on real image features that have no synthetic analogue. The F1-confidence curve collapsing toward zero confidence threshold further confirms that this is not merely a calibration issue and the model lacks the internal representations to make confident detections, not simply a mis calibrated threshold.

The performance hierarchy among integration strategies ($C4 > C5 > C3 > C1$) reveals a clear principle: the proportion of real-image gradient influence during training determines both the ceiling of recall attainable and the final real-domain alignment of the learnt feature space. Condition 4's real dominance (70% of images are real) anchors the model's classification and localisation features to the real-world distribution throughout training; the 30% synthetic contribution expands the pedestrian appearance space without displacing that anchor. Condition 5's sequential design achieves comparable overall performance but via a different

operating profile, higher precision at slightly lower recall, because Stage 1 establishes a tightly localising feature space from synthetic clean annotations, while Stage 2 corrects distributional alignment to real imagery. These two pathways arriving at similar overall mAP50 but with distinct precision–recall operating characteristics suggests that deployment context should inform strategy selection: if minimising false positives is the priority (e.g., avoiding false emergency braking), Condition 5 is preferable; if maximising detection sensitivity is the priority (e.g., urban surveillance where missed pedestrians carry greater cost), Condition 4 is the recommended strategy.

Condition 3's competitiveness (+10.3% mAP50 at equal epoch count) is itself a noteworthy result not widely anticipated. Prior literature rarely evaluated naive mixing at full convergence against a consistently trained baseline; in several works, mixed conditions were trained for fewer epochs than real-only baselines, introducing a confound that overstated the disadvantage of mixing. This study, with identical 50-epoch training across all conditions, demonstrates that naive mixing is not inherently harmful and can produce meaningful improvements however, it simply leaves recall below the optimal 70:30 balance because the synthetic distribution shift cannot be fully overcome through gradient updates when it constitutes 50% of training signal.

5.2 Comparison Against Prior Quantitative Results

Training Dataset	Test Dataset		
	MOT17Det	MOT19Det	CityPersons
COCO	0.636	0.466	0.546
MOT17Det	-	0.605	0.571
MOT19Det	0.618	-	0.419
CityPersons	0.710	0.488	-
ViPeD	0.721	0.629	0.516
ViPeD + Real	0.733	0.582	0.546

Figure 6: Cross-dataset AP@0.5 results from ViPeD [30]

TABLE I
RATIOS OF REAL AND SYNTHETIC DATA FOR THE TRAINING.

EXPERIMENT	TRAINING DATA			
	RealStreet	SynthStreet	SupporterReal	SupporterVirtual
EXP1	100%	0%	0%	0%
EXP2	100%	0%	100%	0%
EXP3	0%	100%	0%	0%
EXP4	0%	100%	0%	100%
EXP5	100%	100%	100%	100%
EXP6	10%	90%	0%	0%
EXP7	20%	80%	0%	0%

TABLE V
PERFORMANCE OF THE EXPERIMENTS FOR ALL CLASSES

EXPERIMENT	mAP@0.5
EXP1	0.629
EXP2	0.594
EXP3	0.305
EXP4	0.436
EXP5	0.768
EXP6	0.509
EXP7	0.638

Figure 7: Training data compositions AND mAP@0.5 results for Experiments EXP1–EXP7 [31]

TABLE 3. Cross-dataset validation results: Assessing the impact of our synthetic dataset (MetaPed) on model performance across different training and testing sets (AP at IoU 0.5 and AR at IoU from 0.50 to 0.95 in large area).

Train Datasets	Test Datasets	AP at IoU 0.5	AR at IoU from 0.50 to 0.95
Citypersons	INRIA Person	0.894	0.649
Citypersons + MetaPed	INRIA Person	0.928	0.680
KITTI	INRIA Person	0.903	0.538
KITTI + MetaPed	INRIA Person	0.971	0.602
INRIA Person	Citypersons	0.231	0.460
INRIA Person + MetaPed	Citypersons	0.286	0.400
KITTI	Citypersons	0.382	0.451
KITTI + MetaPed	Citypersons	0.394	0.470
Citypersons	KITTI	0.543	0.466
Citypersons + MetaPed	KITTI	0.579	0.471
INRIA Person	KITTI	0.485	0.384
INRIA Person + MetaPed	KITTI	0.560	0.454

TABLE 4. Validation results: Impact of our synthetic dataset (MetaPed) on same training and testing sets (AP at IoU 0.5 and AR at IoU from 0.50 to 0.95 in large area).

Train Datasets	Test Datasets	AP at IoU 0.5	AR at IoU from 0.50 to 0.95
INRIA Person	INRIA Person	0.958	0.697
INRIA Person + MetaPed	INRIA Person	0.988	0.725
Citypersons	Citypersons	0.630	0.593
Citypersons + MetaPed	Citypersons	0.639	0.582
KITTI	KITTI	0.745	0.554
KITTI + MetaPed	KITTI	0.752	0.536

TABLE 5. Cross-dataset validation results: Evaluating the effectiveness of our synthetic dataset (MetaPed) on different testing sets (AP at IoU 0.5 and AR at IoU from 0.50 to 0.95 in large area).

Train Datasets	Test Datasets	AP at IoU 0.5	AR at IoU from 0.50 to 0.95
Citypersons	INRIA Person	0.894	0.649
KITTI	INRIA Person	0.903	0.538
MetaPed	INRIA Person	0.953	0.660
INRIA Person	Citypersons	0.231	0.460
KITTI	Citypersons	0.382	0.451
MetaPed	Citypersons	0.367	0.482
Citypersons	KITTI	0.543	0.466
INRIA Person	KITTI	0.485	0.384
MetaPed	KITTI	0.588	0.483

Figure 8: Cross-dataset and intra-dataset AP@0.5 and AR results from MetaPed [25]

Figures 6–8 present results from three prior studies testing synthetic-to-real integration. Direct numerical comparison is not possible given differences in architecture, corpus scale, and benchmark, but directional patterns are informative.

The synthetic-only collapse is consistent across limited-scale studies: Khalzaa et al. report a 51.5% relative drop (EXP3: 0.305 vs. 0.629) using YOLOv7 on a custom synthetic road dataset [31], closely mirroring this study's −62.8% (Condition 2). ViPeD's contrasting result — synthetic-only AP = 0.721 exceeding the real-only baseline of 0.710 — demonstrates that at sufficient scale (~50,000 high-fidelity GTA-V frames) [30], the domain gap can be largely closed without real fine-tuning, setting

the upper bound this project cannot reach at its dataset scale but confirming that the gap is a function of synthetic quality, not an architectural constant.

The apparent inversion in optimal mixing ratios between this study (70% real, 30% synthetic, +12.7%) and Khalzaa's EXP7 (20% real, 80% synthetic, +1.4%) is explained by domain gap severity: when synthetic imagery is geometrically close to the target domain, higher synthetic proportions are tolerable; when the gap is large as between SYNTHIA-PANO's panoramic projection and real urban surveillance footage, real images must dominate to keep the learnt feature space anchored to the evaluation distribution. MetaPed's modest +1.4% gain from synthetic augmentation, despite using a larger and more diverse real corpus than this study, confirms that synthetic benefit scales inversely with the coverage already provided by the real training set; this study's 2,500-image corpus has considerably more headroom, explaining the larger observed gains.

Critically, all three comparison studies use anchor-based architectures, Khalzaa et al. use YOLOv7 while MetaPed and ViPeD use Faster R-CNN [25], [30], [31]. No prior work has applied these strategies to anchor-free YOLOv8n. The consistent gains across all three integration conditions in this study confirm that synthetic augmentation benefits extend to anchor-free single-stage detection under CPU-only constraints, filling the architectural gap identified in Section 2.8.

5.3 Limitations of the Study

Several limitations of this study are acknowledged and should inform the interpretation of its findings.

The most direct limitation is dataset scale. With 2,500 real training images, the baseline and mixed conditions operate well below the scale of leading pedestrian benchmarks such as EuroCity Persons at 238,200 instances [15]. Gains from synthetic data are therefore partly a function of the deficiencies in the real corpus rather than exclusively a property of the integration strategy. Larger, more diverse real datasets may produce smaller marginal gains from the same synthetic additions, as MetaPed's results illustrate [25].

The mixed dataset construction involves a single random sample of 750 synthetic images for Condition 4 and 2,500 for Condition 3. This introduces sampling variance in performance estimates that is not quantified in the current study. It is plausible that

a different random draw would produce modestly different results, and the true expected performance of each strategy across the sampling distribution is not known. A repeat-sampling experiment is identified as priority future work in Section 5.6.

Single-class training is a further limitation that is rarely discussed in comparable studies. Configuring the detection head for a single class for only pedestrian simplifies the classification problem relative to multi-class deployment contexts, where the model must additionally distinguish pedestrians from cyclists, vehicles, and street furniture.

No cross-dataset generalisation testing was conducted. All results are reported on an internal validation split drawn from the same distribution as the real training corpus. Whether the performance gains from synthetic integration persist when the model is evaluated on a genuinely different real-world benchmark such as EuroCity Persons or INRIA Person is untested. Cross-dataset evaluation is standard in the literature and would strengthen the generalisability of this study's conclusions [25], [30].

Finally, the 50-epoch training duration on CPU hardware, while consistent across all conditions and justified by the controlled design objective, does not guarantee full convergence. Condition 4's learning curve had not fully plateaued by epoch 50, suggesting further gains may be achievable with extended training.

5.4 Practical Recommendations

For practitioners working under similar resource constraints, the results suggest a clear practical hierarchy. When compute budget and training time are limited, 70:30 real-dominated mixing offers the strongest performance-to-cost trade-off. When higher bounding-box precision is a priority (for example, in dense traffic scenarios where accurate localisation supports trajectory prediction), sequential pretraining followed by real fine-tuning remains a viable option, despite the marginal recall trade-off and doubled compute cost. Equal-weight 50/50 mixing can work when synthetic data quality is high and sufficient training time is available, but it is generally less efficient than real-dominated mixing.

The main theoretical contribution of this study is the empirical demonstration that synthetic-to-real transfer benefits previously observed with two-stage detectors also apply to the anchor-free, single-stage YOLOv8n architecture. The warm-start

dynamics and consistent elevation of the precision–recall curve suggest that structural pedestrian features learned from synthetic data transfer effectively at the representation level, even when the detection head design differs significantly.

5.5 Conclusions

This project set out to determine whether, and by what integration mechanism, synthetic pedestrian imagery improves the real-world detection performance of YOLOv8n, using a controlled five-condition experiment on a fixed real-domain validation set of 441 images and 2,705 ground-truth instances.

Three principal conclusions are drawn from the evidence:

1. Synthetic data integration measurably improves YOLOv8n's real-world pedestrian detection performance when implemented correctly. The best-performing integration strategy (70:30 real-dominated mixing) achieves $mAP_{50} = 0.666$, a 12.7% relative improvement over the real-only baseline of 0.591, with simultaneous gains in precision (+6.8%), recall (+9.8%), and mAP_{50-95} (+14.4%). These improvements represent an upward displacement of the precision–recall curve rather than a threshold shift, confirming genuine representational improvement attributable to synthetic training diversity.
2. The integration mechanism determines both the magnitude and the character of the performance gain. 70:30 real-dominated mixing achieves the highest overall mAP_{50} and recall; sequential synthetic pretraining followed by real fine-tuning achieves the highest precision via a different mechanism, synthetically transferred bounding-box localisation quality. Both outperform naive 50/50 mixing, which still improves over the real-only baseline but constrains recall due to the heavier distributional dilution imposed by an equal synthetic proportion.
3. The patterns observed for YOLOv8n are directionally consistent with prior evidence from Faster R-CNN-based literature but cannot be inferred from that literature alone. The warm-start transfer dynamic, the domain gap characterisation, and the performance hierarchy across integration strategies all replicate findings from MetaPed, ViPeD, and MOTSynth, confirming that these phenomena generalise to anchor-free single-stage architectures [18], [25], [30]. However, the specific precision advantage of sequential pretraining,

the warm-start mAP50 value of 0.409 at Stage 2 epoch 1, and the recall profile differences between conditions are architecture-specific empirical contributions not available in existing literature.

Together, these findings provide controlled evidence that synthetic-to-real pedestrian transfer strategies validated for Faster R-CNN are also effective for YOLOv8n, extending the practical applicability of this body of knowledge to one of the most widely deployed modern detection frameworks under CPU-constrained edge-deployment conditions.

5.6 Future Work

Several concrete directions would build on these findings. A finer-grained sweep of mixing ratios (10–70% synthetic) would map the performance curve more precisely and help identify any optimal inflection point. Cross-dataset evaluation on established benchmarks such as CityPersons and EuroCity Persons would test the generalisation of the observed gains. Given the inclusion of SYNTHIA-PANO, a dedicated panoramic real validation set would help isolate the contribution of wide field-of-view training data. Further improvements could be explored through photorealistic texture transfer or advanced domain randomisation to reduce the distributional gap. Finally, deploying the best-performing model (Condition 4) on edge hardware such as a Jetson Nano or Raspberry Pi 5 would provide a complete feasibility assessment for real-world embedded autonomous driving applications.

6 References

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7 Appendix

7.1 A: Augmentation Pipeline Parameters

```
imranmohamed@Imrans-MacBook-Air yolo_dir_ped % cd ~/Downloads/real_synth_5050

yolo detect train \
  model=yolov8n.pt \
  data=data.yaml \
  imgsz=640 \
  epochs=50 \
  batch=4 \
  auto_augment=randaugment \
  hsv_h=0.015 hsv_s=0.7 hsv_v=0.4 \
  flipplr=0.5 flipud=0.0 \
  erasing=0.4 \
  mosaic=1.0 mixup=0.0 \
  translate=0.1 scale=0.5 shear=0.0 perspective=0.0 \
  name=real_synth_5050_50ep_small150
```

Augmentation	Parameter	Rationale
HSV colour jitter	hue ± 0.015 , saturation ± 0.7 , value ± 0.4	Reduces sensitivity to synthetic colour palette and lighting artefacts
Horizontal flip	p = 0.5	Simulates symmetric pedestrian scenarios
Random erasing	p = 0.4	Encourages part-based detection; mitigates lack of occlusion diversity in synthetic data
Mosaic augmentation	enabled; disabled final 10 epochs (<code>close_mosaic=10</code>)	Increases instances-per-batch; disabling near convergence stabilises loss
Auto-augment policy	RandAugment	Stochastic additional transforms beyond explicit parameters
Scale jitter	± 0.5	Pedestrian size variability
Translation jitter	± 0.1	Spatial position variability

7.2 B: Validation Run Output Snippets for Final YOLOv8n Experiments

```
Validating /Users/[redacted]/Downloads/yolo_dir_ped/runs/detect/real_only_50ep/weights/best.pt...
Ultralytics 8.3.249 Python-3.9.6 torch-2.8.0 CPU (Apple M2)
Model summary (fused): 72 layers, 3,085,843 parameters, 0 gradients, 8.1 GFLOPs
Class Images Instances Box(P R mAP50 mAP50-95): 100% 56/56 2.1it/s 26.7s
all 441 3157 0.693 0.513 0.591 0.375

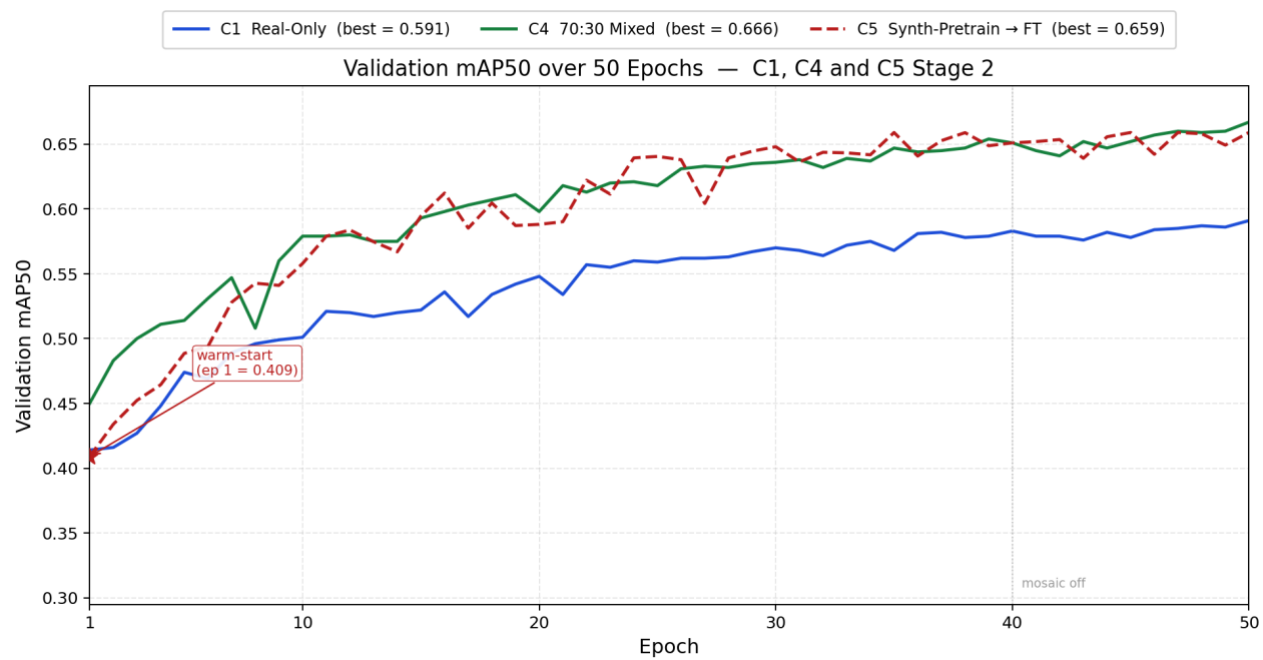
yolo detect val \
model=/Users/[redacted]/Downloads/synth_combo/runs/detect/synth_combo_50ep_small150/weights/best.pt \
data=data.yaml \
imgsz=640 \
name=synth_combo_to_real_50ep_small150
Ultralytics 8.3.249 Python-3.9.6 torch-2.8.0 CPU (Apple M2)
Model summary (fused): 72 layers, 3,085,843 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access (nino: 0.0±0.0 ms, read: 1396.8±24.2 MB/s, size: 2331.6 KB)
val: Scanning /Users/[redacted]/Downloads/yolo_dir_ped/valid/labels.cache... 441 images, 59 backgrounds, 0 corrupt: 100% 441/441 8.8Mit/s 0.0s
Class Images Instances Box(P R mAP50 mAP50-95): 100% 28/28 1.7s/it 48.8s
all 441 2785 0.227 0.241 0.22 0.186

yolo detect val \
model=/Users/[redacted]/Downloads/real_synth_50ep/runs/detect/real_synth_50ep_small150/weights/best.pt \
data=data.yaml \
imgsz=640 \
name=real_synth_50ep_to_real_50ep_small150
Ultralytics 8.3.249 Python-3.9.6 torch-2.8.0 CPU (Apple M2)
Model summary (fused): 72 layers, 3,085,843 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access (nino: 0.0±0.0 ms, read: 1369.1±24.8 MB/s, size: 2384.6 KB)
val: Scanning /Users/[redacted]/Downloads/yolo_dir_ped/valid/labels.cache... 441 images, 59 backgrounds, 0 corrupt: 100% 441/441 6.1Mit/s 0.0s
Class Images Instances Box(P R mAP50 mAP50-95): 100% 28/28 1.7s/it 48.3s
all 441 2785 0.745 0.549 0.652 0.42

Learn more at https://docs.ultralytics.com/modes/val
yolo_dir_ped % yolo detect val model=/Users/[redacted]/Downloads/real_plus_syn30/runs/detect/real_plus_syn30_50ep_small150/weights/best.pt data=data.yaml imgsz=640 plots=True
Ultralytics 8.3.249 Python-3.9.6 torch-2.8.0 CPU (Apple M2)
Model summary (fused): 72 layers, 3,085,843 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access (nino: 0.0±0.0 ms, read: 1150.3±147.8 MB/s, size: 2330.3 KB)
val: Scanning /Users/[redacted]/Downloads/yolo_dir_ped/valid/labels.cache... 441 images, 59 backgrounds, 0 corrupt: 100% 441/441 5.4Mit/s 0.0s
Class Images Instances Box(P R mAP50 mAP50-95): 100% 28/28 2.1s/it 59.1s
all 441 2785 0.74 0.563 0.666 0.429
Speed: 1.0ms preprocess, 89.4ms inference, 0.0ms loss, 0.3ms postprocess per image
Results saved to /Users/[redacted]/Downloads/yolo_dir_ped/runs/detect/7898_mixed_full

yolo_dir_ped % yolo detect val \
p_dir_ped/runs/detect/finetune_synth_combo_to_real_50ep/weights/best.pt" \
imgsz=640 \
batch=4
Ultralytics 8.3.249 Python-3.9.6 torch-2.8.0 CPU (Apple M2)
Model summary (fused): 72 layers, 3,085,843 parameters, 0 gradients, 8.1 GFLOPs
val: Fast image access (nino: 0.0±0.0 ms, read: 1238.5±159.0 MB/s, size: 2323.0 KB)
val: Scanning /Users/[redacted]/Downloads/yolo_dir_ped/valid/labels.cache... 441 images, 59 backgrounds, 0 corrupt: 100% 441/441 11.3Mit/s 0.0s
Class Images Instances Box(P R mAP50 mAP50-95): 100% 111/111 4.4it/s 25.2s
all 441 2785 0.751 0.558 0.659 0.423
Speed: 0.2ms preprocess, 22.4ms inference, 0.0ms loss, 0.2ms postprocess per image
Results saved to /Users/[redacted]/Downloads/yolo_dir_ped/runs/detect/val3
```

7.3 C: Training Dynamics — Per-Epoch Validation mAP50 Learning Curves



7.4 Monthly Report Forms

Monthly Report Form 2025-26

This form is meant to record the status of the project work in three key points: the end of December 2025, the end of January 2026, and the end of February 2026.

Instructions for student: *please complete the fields in the table below. Then send the form to your supervisor for comments and feedback. Once you receive the final form from your supervisor, please keep it in your portfolio and append it to your final report. You do not need to submit the form to anyone else.*

Student name	Redacted	Student number	Redacted
Supervisor name	Redacted		
Month	December <i>(please delete as appropriate)</i>		

Self-assessment of project: student's own critical review of work done so far	
To be completed by the student. <i>Please describe and assess your progress with respect to your project plan. Explain any problems.</i>	Finalized project scope, picked datasets and formulated research around pedestrian detection in autonomous vehicles and investigated models used, and chosen model to use in experiments
Key objectives and milestones for the next period	
To be completed by the student. <i>Please provide your planned work and priorities for the next period. Explain what you expect to have accomplished in the next few weeks.</i>	To complete training and validation runs of yolov8n on 2 chosen datasets and record the investigative data received.
Supervisor feedback	
To be completed by the supervisor. <i>Please write comments regarding current progress and workplan for the work ahead. Then return the form to the student.</i>	Good initial progress with clear scope and model selection. Next, focus on experiments, evaluation, and documentation. Overall satisfactory progress.

Monthly Report Form 2025-26

This form is meant to record the status of the project work in three key points: the end of December 2025, the end of January 2026, and the end of February 2026.

Instructions for student: *please complete the fields in the table below. Then send the form to your supervisor for comments and feedback. Once you receive the final form from your supervisor, please keep it in your portfolio and append it to your final report. You do not need to submit the form to anyone else.*

Student name	Redacted	Student number	Redacted
Supervisor name	Redacted		
Month	January <i>(please delete as appropriate)</i>		

Self-assessment of project: student's own critical review of work done so far	
To be completed by the student. <i>Please describe and assess your progress with respect to your project plan. Explain any problems.</i>	Finalized 5 conditions for training to be completed upon with the new idea of mixing real and synthetic datasets. The training has begun and been completed for real only baseline and synthetic only.
Key objectives and milestones for the next period	
To be completed by the student. <i>Please provide your planned work and priorities for the next period. Explain what you expect to have accomplished in the next few weeks.</i>	The following runs of a 50/50 mix, 70/30 mix and fine tune condition are yet to be completed. Must be completed within the month and data to be recorded in a comparative manner. Report writing to be started once all data received.
Supervisor feedback	
To be completed by the supervisor. <i>Please write comments regarding current progress and workplan for the work ahead. Then return the form to the student.</i>	Good progress with experiments using mixed datasets. The planned next steps are appropriate. Focus on completing experiments on time and ensuring consistent evaluation.

Monthly Report Form 2025-26

This form is meant to record the status of the project work in three key points: the end of December 2025, the end of January 2026, and the end of February 2026.

Instructions for student: please complete the fields in the table below. Then send the form to your supervisor for comments and feedback. Once you receive the final form from your supervisor, please keep it in your portfolio and append it to your final report. You do not need to submit the form to anyone else.

Student name	Redacted	Student number	Redacted
Supervisor name	Redacted		
Month	February <i>(please delete as appropriate)</i>		

Self-assessment of project: student's own critical review of work done so far	
To be completed by the student. <i>Please describe and assess your progress with respect to your project plan. Explain any problems.</i>	Completed all investigation of condition and now onto report writing. In terms of project plan, on track and going through methodology section
Key objectives and milestones for the next period	
To be completed by the student. <i>Please provide your planned work and priorities for the next period. Explain what you expect to have accomplished in the next few weeks.</i>	To keep on pushing through the report writing stages and to have a working draft by mid-march. Report plan is already finalized, just need to execute and fully expand on all sections.
Supervisor feedback	
To be completed by the supervisor. <i>Please write comments regarding current progress and workplan for the work ahead. Then return the form to the student.</i>	Good progress with completion of experiments and transition to report writing. The plan is appropriate. Focus on producing a clear, well-structured draft with critical analysis of results. Overall, on track.